



Physical, mechanical and structural properties of BaO–CaF₂–P₂O₅ glasses



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ABSTRACT

Mechanical properties and structure of melt-quenched ternary barium phosphate glasses belonging to the composition $(50 - X)\text{BaO}-X\text{CaF}_2-50\text{P}_2\text{O}_5$ ($X = 0$ to 10 mol%) were studied using Vickers indentation and vibrational spectroscopy. Elastic moduli of prepared samples were calculated using the Makishima and Mackenzie (M–M) model. Density of glasses was measured using the Archimedes method, while molar volume and atomic packing density were calculated from measured density to study structural changes in the glass matrix. Vickers hardness and brittleness increased and fracture toughness decreased with increase in CaF₂ content. This can be attributed to increase in the compactness of phosphate bonds and reduction in the molar volume of glasses with partial substitution of Ba²⁺ with Ca²⁺. Infrared and Raman spectroscopies revealed that the glass network mainly consisted of metaphosphate units arranged in chain and ring structures together with small amounts of ultraphosphate and pyrophosphate units.

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1. Introduction

Phosphate glasses have superior properties such as high refractive index, high ultraviolet (UV) transmission, high electrical conductivity, low melting temperature and low glass transition temperature, which make them suitable candidates for technological applications such as optical fibers for data transmission, host glasses for solid state lasers, luminescent solar energy concentrators, immobilization of nuclear wastes, solid state batteries and glass-to-metal sealing [1,2,3]. Poor thermal stability against crystallization, low chemical durability, volatile and hygroscopic nature of phosphate glasses limit their extensive practical applications. These properties can be improved by introducing different metal oxides into the glass matrix [4,5]. Barium phosphate glasses with higher concentration of barium oxide (≥ 50 mol%) show high solubility of active ions, low photoelastic constants and good water durability compared to other alkali earth phosphate glasses [6]. Barium and calcium metaphosphate glasses are optically transparent dielectric matrices used for the preparation of noble metal nanoparticle embedded glasses due to their high metal solubility [7]. Addition of calcium fluoride (CaF₂) to the glass batch reduces the liquidus temperature to a substantial extent and improves the durability of glasses by reducing the hydroxyl content [8,9].

Glasses with excellent mechanical and elastic properties are required for many applications, which include glass components for television tubes, architectural use, optical fibers for data transmission, temporary implants for bone repair and reconstruction [10,11,12].

Therefore, an investigation of both mechanical and elastic properties is necessary for the selection of appropriate glasses for technological and biomedical applications. Properties of all glasses, including physical and mechanical properties depend on their local structure determined by the composition [13]. Structure determination in glasses is a difficult task compared to that of a crystal, because there is no particular method available for investigating the structure of an amorphous solid [14]. Some perception of the glass structure can be obtained by combining results of different spectroscopic techniques such as infrared, Raman, NMR, neutron and X-ray diffraction. Raman and infrared spectroscopies are two powerful tools, which exhibit dependence of local structure on composition and help to understand the effect of composition on glass properties [15]. Hence both these techniques are used to study short and intermediate range orders in glass. Density of glasses, which is related to molar volume and atomic packing density, also plays an important role in the study of structure of glasses [16].

In the previous paper [8], we have reported that the addition of calcium fluoride (CaF₂) to barium metaphosphate glasses improves their thermal stability against crystallization and glass forming ability. These glasses with composition $(50 - X)\text{BaO}-X\text{CaF}_2-50\text{P}_2\text{O}_5$ (BaPF, $X = 0$ to 10 mol%) also show thermal stability parameter of 100 °C, hence they are suitable for fiber fabrication. In the present work less investigated properties like brittleness, fracture toughness and hardness of these glasses are studied using Vickers indentation and Makishima and Mackenzie model (1973 and 1975) to check the improvement in mechanical properties with CaF₂ addition. Change in density, molar volume and atomic packing density with composition are also being investigated to get an insight into the structure of prepared glasses. In addition to this, infrared and Raman spectroscopies are used to probe the interaction of cations with their local environment and to determine

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the types of the phosphate tetrahedra present in the glass matrix. Efforts are also made to correlate the physical and mechanical properties with structure, which will help to design better glasses with superior properties.

2. Materials and methods

2.1. Sample preparation

Glasses were prepared by melt-quenching process using high purity barium carbonate (BaCO_3 with 99.8% purity), calcium fluoride (CaF_2 with 99%) and ammonium dihydrogen phosphate ($\text{NH}_4\text{H}_2\text{PO}_4$ with 98%) supplied by Alfa Aesar. A 20 g batch of each composition was weighed, mixed well in an agate mortar for 30 min and then calcined at 400 °C for 2 h in a muffle furnace to remove ammonia and water. Calcined batch was taken in a silica crucible and melted in a PID controlled high temperature furnace at 1200 °C. The melt was kept at that temperature for 1 h to get the required homogeneity. The homogenized melt was quenched in air on copper plates at room temperature, annealed immediately at 400 °C for 3 h to remove the residual thermal stresses developed during the quenching and then furnace cooled to room temperature. Prepared samples were ground and polished using emery papers of different grit size (120–2000) and alumina powder (0.05 μm) suspension to get samples with flat surfaces suited for measuring mechanical properties. Batch composition of glasses used in this study is given in Table 1.

2.2. Sample characterizations

X-ray diffraction (XRD) patterns of powdered samples were obtained using JEOL JDX-8P-X-ray diffractometer with $\text{Cu-K}\alpha$ radiation at a scanning rate of 4°/min in the range of 10°–80° diffraction angle (2θ). The X-ray tube was operated at 40 kV and 15 mA. Density (ρ) of bulk glass samples was obtained by the Archimedes method at room temperature using xylene as the immersion liquid. Measurements were carried out for three specimens of each composition and average values of density were obtained. Vickers hardness (H_v) of samples at room temperature was measured using the Vickers diamond pyramid indentation technique (Clemex microhardness tester, MMT X7, Matsuzawa Seiki Corp., Japan) by applying a load of 0.98 N for 10 s. Measurements were repeated ten times for each sample and mean values of hardness were calculated. Fracture toughness studies were carried out by applying a load of 9.8 N to the samples. Radial-median type cracks were observed and the length of crack was measured using an optical microscope. Vickers indentations made on the surface of the glasses were further analyzed using scanning electron microscope (SEM, JEOL JSM 6380LA system, Japan). All samples were gold sputtered using a JFC 1600 auto fine coater (JEOL, Japan) before carrying out the SEM analysis to avoid charging of the samples. Energy dispersive analysis of X-ray (EDS) was also carried out using SEM to determine the elemental composition of the glasses. Fourier transform infrared (FTIR) spectra of samples were recorded in the wavenumber range 400–4000 cm^{-1} in the transmission mode of the FTIR spectrometer (Jasco FTIR-4200, Japan) for an average of 32 scans using the standard KBr pellet method. Micro Raman spectra of fine powdered glass samples were collected in

the wavenumber range 400–2000 cm^{-1} using a Laser Confocal Raman microscope (inVia, Renishaw, UK) equipped with an argon ion laser source of 100 mW power and 514 nm wavelength.

3. Results

3.1. Physical properties

X-ray diffraction (XRD) patterns of powdered samples obtained between diffraction angle (2θ) 10°–80° are shown in Fig. 1. Presence of a broad feature (amorphous halo) between 2θ angle 20°–30° and absence of sharp peaks in the pattern confirm the amorphous nature of the prepared samples.

Density of samples was measured using the Archimedes method. From measured values of density (ρ), molar volume (V_m) was calculated using the following equation

$$V_m = \frac{M}{\rho} \quad (1)$$

where M is the molecular weight of glass samples calculated from batch composition.

Atomic packing density (V_t), defined as the ratio of minimum theoretical volume occupied by the ions to the effective volume of the glass [17] is given by

$$V_t = \rho \frac{\sum_i V_i X_i}{\sum_i M_i X_i} \quad (2)$$

where X_i and M_i are the mole fraction and molecular weight of the component i , respectively. V_i is the volume that ions occupy in the component i , which can be formulated for a compound $M_p N_q$ as

$$V_i = \frac{4}{3} \pi (pR_M^3 + qR_N^3) N_A \quad (3)$$

where $M_p N_q$ is a metal oxide or metal fluoride having ions M and N with radii R_M and R_N . N_A is the Avogadro's number. In this paper, V_i values of individual compounds needed for the calculation of atomic packing density (V_t) were selected from the reference [17].

Measured density of bulk glasses and calculated values of molar volume and atomic packing density are given in the Table 1. Assuming that uncertainty in density measurements are independent and random, error in measured value of density is evaluated by taking the standard deviation of the mean (SDOM) of the obtained density values for three specimens of each composition. Taking uncertainty in measured density, error in molar volume and atomic packing density are calculated using quadrature sum of uncertainty rule of error propagation. All the errors reported in this manuscript are rounded to one significant figure. As shown in Table 1, the density of the glasses decreases, molar volume remains almost constant but with a decreasing trend and atomic packing density shows a slight increment with increase in CaF_2 mol%.

Table 1

Batch composition, density (ρ), molar volume (V_m) and atomic packing density (V_t) of BaPF glasses. Error is given as the standard deviation of the mean (SDOM).

Sample code	P_2O_5 (mol%)	BaO (mol%)	CaF_2 (mol%)	ρ (g/cm^3)	V_m (cm^3)	V_t
BaPF0	50	50	0	3.541 ± 0.004	41.68 ± 0.05	0.574 ± 0.001
BaPF2	50	48	2	3.508 ± 0.005	41.65 ± 0.06	0.575 ± 0.001
BaPF4	50	46	4	3.472 ± 0.005	41.65 ± 0.06	0.576 ± 0.001
BaPF6	50	44	6	3.438 ± 0.005	41.62 ± 0.06	0.576 ± 0.001
BaPF8	50	42	8	3.402 ± 0.006	41.62 ± 0.07	0.577 ± 0.001
BaPF10	50	40	10	3.365 ± 0.005	41.62 ± 0.06	0.577 ± 0.001

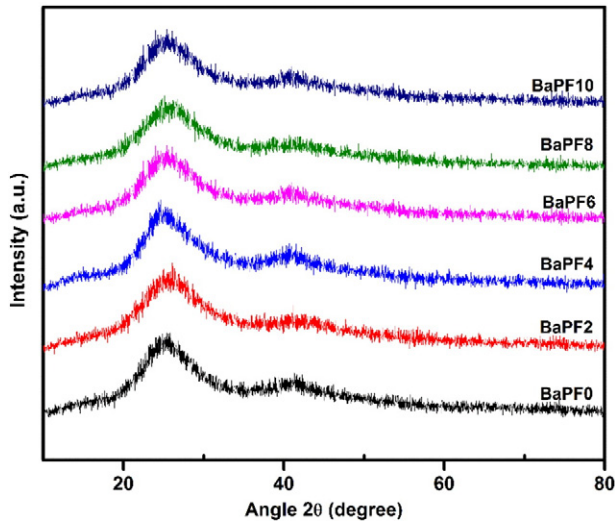


Fig. 1. XRD patterns of $(50 - X)\text{BaO}-X\text{CaF}_2-50\text{P}_2\text{O}_5$ (BaPF) glasses.

3.2. Mechanical properties

Makishima and Mackenzie [18,19] calculated elastic moduli of oxide glasses using dissociation energy of oxide constituents per unit volume (G_i) and packing density (V_t). Inaba et al. [20] carried out an extensive study of Young's modulus of oxide glasses including phosphate glasses and modified dissociation energy (G_i) of P_2O_5 from 62.8 kJ/cm^3 to 28.28 kJ/cm^3 by considering $\text{P} = \text{O}$ to be a non-bridging bond, without any contribution to stiffness. This modification of dissociation energy helped to predict Young's modulus (E) of phosphate glasses with more accuracy. Hager [21] and Akhilesh et al. [17] proposed that the Makishima and Mackenzie (M-M) model can be used to calculate elastic moduli of oxyfluoride glasses. All the above mentioned facts indicate that elastic moduli of both phosphate and oxyfluorophosphate glasses can be obtained using the Makishima and Mackenzie model. So in the present study elastic moduli of the $(50 - X)\text{BaO}-X\text{CaF}_2-50\text{P}_2\text{O}_5$ glasses were calculated using the theoretical model proposed by Makishima and Mackenzie [18, 19] with modified dissociation energy values given by Inaba et al. [20]. The M-M model defines Young's modulus of polycrystalline glasses as

$$E = 83.6V_t \sum_i G_i X_i \quad (4)$$

where V_t is the packing density as defined by Eq. (2), G_i is the dissociation energy of oxide or fluoride constituents per unit volume and X_i is the mole fraction of the component i . Young's modulus of the prepared samples was calculated using Eqs. (2) and (4), taking the values of G_i and V_t of individual components from reference [17]. Poisson's ratio (σ) of the glasses was determined using the equation proposed by Makishima and Mackenzie [19]

$$\sigma = 0.5 - \frac{1}{7.2V_t} \quad (5)$$

which directly relates Poisson's ratio to packing density (V_t). From Young's modulus, Poisson's ratio and packing density, shear modulus (S) and bulk modulus (K) were calculated using the following relations

$$S = \frac{E}{2(1 + \sigma)} \quad (6)$$

$$K = 1.2V_t E. \quad (7)$$

Values of elastic moduli calculated using the above equations are listed in Table 2. Errors are not shown for these values, since all are

Table 2

Young's modulus (E), bulk modulus (K), shear modulus (S) and Poisson's ratio (σ) of BaPF glasses.

Sample code	E (GPa)	K (GPa)	S (GPa)	σ
BaPF0	38.63	26.61	15.35	0.258
BaPF2	39.24	27.08	15.59	0.258
BaPF4	39.86	27.55	15.83	0.259
BaPF6	40.41	27.93	16.05	0.259
BaPF8	41.03	28.41	16.29	0.259
BaPF10	41.51	28.74	16.49	0.259

theoretically calculated. The values of parameters listed in Table 2 lie within the range of values reported for phosphate glasses [22].

Mechanical properties of BaPF glasses such as Vickers hardness, fracture toughness and brittleness are given in Table 3. Error in measured hardness is obtained by evaluating standard deviation of the mean (SDOM) of hardness values obtained across ten indentations. Similarly error in measured crack length also obtained by calculating SDOM of crack lengths across ten indentations. Error in fracture toughness and brittleness are obtained from measured results of hardness and crack length by applying rule of error propagation. Vickers hardness and brittleness increase whereas the fracture toughness of the samples decreases as BaO content decreases.

Strength of glasses mainly depends upon two factors, crack initiation and crack propagation. Crack propagation can be evaluated by measuring fracture toughness (K_{IC}), which defines the ease with which a crack can propagate in a material. Fracture toughness (K_{IC}) of the glasses is usually calculated using an equation developed by Anstis et al. [23], which involves the direct crack measurement method. This can be formulated as

$$K_{IC} = 0.016 \left(\frac{E}{H} \right)^{1/2} \left(\frac{P}{C^{3/2}} \right) \quad (8)$$

where K_{IC} is the fracture toughness in $\text{MPa(m)}^{1/2}$, E , H , P and C are calculated Young's modulus (GPa), measured Vickers hardness (GPa), peak load applied (9.8 N) and measured half crack length (μm) respectively. Young's modulus values obtained using the Makishima and Mackenzie model are used for the calculation of fracture toughness of the samples. Crack initiation in glasses mainly depends on their brittleness [24]. Brittleness or simple index of brittleness (B) of glasses can be calculated from the method proposed by Lawn and Marshall [25], and is given by the ratio of Vickers hardness (resistance to deformation, H_v) to fracture toughness (resistance to fracture, K_{IC})

$$B = \frac{H_v}{K_{IC}} \quad (9)$$

SEM micrographs of Vickers indentations made on the surface of BaPF0 and BaPF10 glasses by applying a load of 9.8 N and EDS spectra showing elemental distribution on the surface of BaPF0 and BaPF10 are given in Fig. 2. In Fig. 2 (a) arrow 1 and arrow 2 at top right show

Table 3

Vickers hardness, half crack length, fracture toughness and brittleness of BaPF glasses. Error is given as the standard deviation of the mean (SDOM).

Sample code	Measured Vickers hardness (GPa)	Measured Half crack length (μm)	Fracture toughness ($\text{MPa(m)}^{1/2}$)	Brittleness ($\mu\text{m}^{-1/2}$)
BaPF0	2.75 ± 0.04	147 ± 1	0.330 ± 0.004	8.34 ± 0.02
BaPF2	2.78 ± 0.04	146 ± 2	0.334 ± 0.007	8.33 ± 0.03
BaPF4	2.80 ± 0.06	150 ± 2	0.322 ± 0.007	8.69 ± 0.03
BaPF6	2.88 ± 0.05	151 ± 2	0.317 ± 0.007	9.10 ± 0.03
BaPF8	2.90 ± 0.03	157 ± 2	0.300 ± 0.006	9.67 ± 0.02
BaPF10	2.92 ± 0.03	157 ± 1	0.301 ± 0.003	9.72 ± 0.01

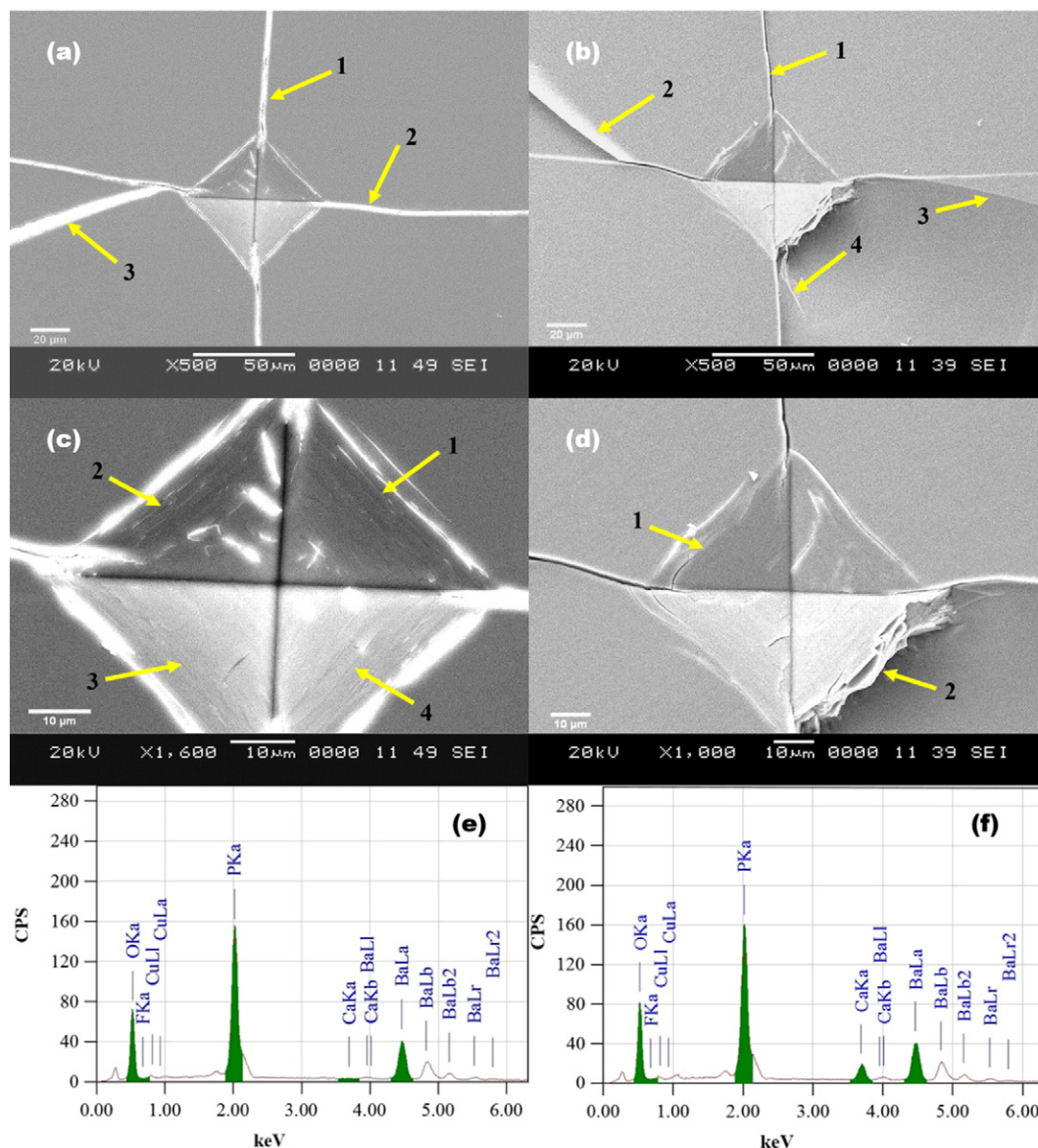


Fig. 2. SEM micrographs of Vickers indentation on (a) BaPF0, (b) BaPF10, (c) magnified view of BaPF0, (d) magnified view of BaPF10 and EDS spectra of (e) BaPF0, (f) BaPF10 glasses.

that cracks formed on the surface due to Vickers indentation grow outward from the four corners of the indentation (median-radial cracks) and arrow 3 at the bottom left indicates the presence of secondary cracks. Arrow 1 at top right in Fig. 2 (b) also shows median-radial crack extending from the impression corner, arrow 2 at top left shows secondary crack originating from the median-radial crack and bottom right arrows (3&4), indicate the presence of shallow lateral crack. Arrows 1, 2, 3 and 4 in Fig. 2 (c) and arrow 1 at top left in Fig. 2 (d) depict the existence of shear faults in the indentation patterns of both BaPF0 and BaPF10, which indicate the occurrence of plastic flow in the samples. In addition to that, bottom right arrow, arrow 2 in Fig. 2 (d) shows a region formed due to the surface chipping of material from the closely spaced shear faults, which indicates increase in the brittleness of the glass. Fig. 2 (e) shows the presence of oxygen, phosphorous and barium elements in BaPF0 and Fig. 2 (f) shows the presence of calcium in addition to all the above mentioned elements in BaPF0. Table 4 shows the mass percentage of elements calculated from batch composition and obtained from EDS analysis for BaPF0 and BaPF10 glasses. In EDS analysis, both spot and area analysis have been carried out to check the elemental composition of the samples. Minimum and

maximum values of mass percentage of elements obtained by carrying out EDS analysis at five different regions of samples are reported in Table 4.

3.3. Structural investigation

The basic structural unit of all phosphates, either crystalline or amorphous is PO_4 tetrahedron. Generally, Phosphorus tetrahedra are represented using Q^i terminology, where i refers to the number of bridging oxygen per tetrahedron. The classification of tetrahedra using Q^i terminology is shown schematically in Fig. 3. Glasses with Q^3 as major phosphate tetrahedral units are known as ultraphosphate glasses, while metaphosphate glasses consist of polymer like glass network formed mainly by the linkage of Q^2 units and invert glasses are made up of pyrophosphate (Q^1) and orthophosphate (Q^0) units [1]. $(50 - X)\text{BaO} - X\text{CaF}_2 - 50\text{P}_2\text{O}_5$ (BaPF) glasses of the present study are expected to consist of polymer like glass network formed by linked Q^2 phosphate tetrahedral units. Ba^{2+} and Ca^{2+} ions are incorporated into the network as glass modifiers, which will act as local charge compensators. As mentioned in the introduction, infrared and Raman spectroscopy are used in

Table 4

Mass percentage of elements calculated from batch composition and obtained from EDS analysis for BaPF0 and BaPF10 glasses. Minimum and maximum values are reported for EDS.

Sample code	Mass percentage of elements							
	Oxygen		Phosphorus		Barium		Calcium	
	Batch composition	EDS	Batch composition	EDS	Batch composition	EDS	Batch composition	EDS
BaPF0	32.51	29.53–30.53	20.98	23.30–23.55	46.52	44.12–45.35	–	–
BaPF10	33.13	32.51–32.83	22.11	24.97–25.68	39.21	37.91–38.05	2.86	2.79–3.13

this work to study the interaction of cations with their local environment and to determine the types of the phosphate tetrahedra present in the glass matrix.

3.3.1. Fourier transform infrared (FTIR) spectroscopy studies

FTIR spectra of BaPF glasses were recorded in the wave number range 400 cm^{-1} to 1400 cm^{-1} and plots were stacked to compare the effect of substitution of BaO by CaF_2 in the batch composition and are shown in Fig. 4. The spectra contain six main stretching vibrations associated with metaphosphate glasses as shown in Table 5. Absorption peaks and line shape of the FTIR spectra of BaPF glasses did not change much as BaO was replaced by CaF_2 in the batch composition. This may be due to both Ba^{2+} and Ca^{2+} ions playing a similar role of network modifiers in the glass matrix. Stretching and bending vibrations of structural units made up of phosphate tetrahedra result in the appearance of bands in mid infrared region (above 300 cm^{-1}).

In Fig. 4, the intense absorption band at $\sim 1259\text{ cm}^{-1}$ is due to asymmetric stretching vibrations of terminal PO_2 groups in Q^2 units, $(\text{PO}_2)_{\text{as}}$, together with some contribution from $\text{P}=\text{O}$ bond stretching mode of Q^3 units, $(\text{P}=\text{O})$. A weak shoulder at $\sim 1153\text{ cm}^{-1}$ is attributed to the weak infrared activation of PO_2 symmetric stretching vibrations associated with Q^2 units, $(\text{PO}_2)_{\text{s}}$, which is the strongest band active in Raman spectrum (Fig. 5). A strong band at $\sim 1108\text{ cm}^{-1}$ is due to the asymmetric stretching vibrations of non-bridging oxygen atoms of Q^1 units terminating metaphosphate chains, $(\text{PO}_3^-)_{\text{as}}$. The band at $\sim 885\text{ cm}^{-1}$ may be due to asymmetric stretching vibrations of $\text{P}-\text{O}-\text{P}$ bridges of Q^2 units in chain and ring type formation, $(\text{P}-\text{O}-\text{P})_{\text{as}}$, while symmetric stretching vibrations of $\text{P}-\text{O}-\text{P}$ linkages associated with linear metaphosphate chains, $(\text{P}-\text{O}-\text{P})_{\text{s}}$, result in bands at ~ 766 and $\sim 710\text{ cm}^{-1}$. Absorption band at $\sim 532\text{ cm}^{-1}$ is due to motion of cations against PO_4^{3-} and band around $\sim 470\text{ cm}^{-1}$ is attributed to the bending vibrations of $\text{O}-\text{P}-\text{O}$ units, $\delta(\text{PO})$ of $(\text{PO}_2)_n$ chains [15,26,27,28,29].

3.3.2. Raman spectroscopy studies

Raman spectra of BaPF glasses in the wavenumber range 400 to 1400 cm^{-1} are shown in Fig. 5. The spectra obtained are similar to the previously reported Raman spectra of barium metaphosphate glasses [4,15,26].

Raman spectra of phosphate glasses usually consist of three spectral regions, i.e. around $950\text{--}1300\text{ cm}^{-1}$, $700\text{--}900\text{ cm}^{-1}$ and below 600 cm^{-1} [30]. Bands around $950\text{--}1300\text{ cm}^{-1}$ region correspond to stretching modes of non-bridging oxygen, while stretching vibrations of bridging oxygen are associated with the bands at $700\text{--}900\text{ cm}^{-1}$ region. The spectral region below 600 cm^{-1} consists of bands due to

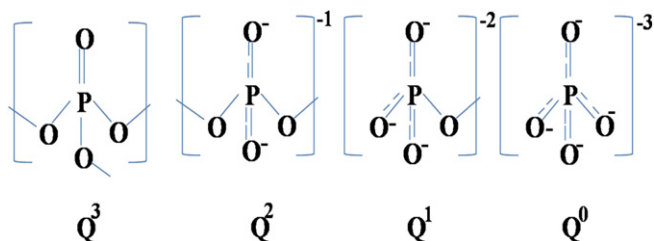


Fig. 3. Schematic illustration of phosphorus tetrahedral sites in phosphate glasses.

complicated internal vibrations like deformation vibrations of phosphate chains and PO_3 deformation vibrations of pyrophosphate segments [26].

The main Raman scattering band situated at $\sim 1157\text{ cm}^{-1}$ shown in Fig. 5 is attributed to the symmetric stretching mode of the two non-bridging oxygen atoms bonded to phosphorus atom, $(\text{PO}_2)_{\text{s}}$, which indicates the formation of Q^2 phosphate tetrahedra. The small width of the band indicates the localized nature of this vibration [26,29,31]. The wavenumber of this band slightly increases from 1157 cm^{-1} to 1164 cm^{-1} as CaF_2 content is increased. The second strongest band observed at $\sim 686\text{ cm}^{-1}$ is due to symmetric stretching vibrations of $\text{P}-\text{O}-\text{P}$ bridges of Q^2 units in the metaphosphate chain, $(\text{P}-\text{O}-\text{P})_{\text{s}}$ [15]. As the CaF_2 content increases, the wavenumber of this band increases to 691 cm^{-1} . The broad band at $\sim 1260\text{ cm}^{-1}$ is assigned to asymmetric stretching mode of the two non-bridging oxygen atoms bonded to phosphorus atom, $(\text{PO}_2)_{\text{as}}$, which shift to 1258 cm^{-1} with increase in CaF_2 content [4,32]. Band at $\sim 518\text{ cm}^{-1}$ is ascribed to bending mode related to cation motion and chain structure [32,33]. Two very weak bands situated at $\sim 1085\text{ cm}^{-1}$ and $\sim 1012\text{ cm}^{-1}$ are due to the asymmetric and symmetric stretching vibrations of non-bridging oxygen atoms of Q^1 units terminating metaphosphate chains, $(\text{PO}_3^-)_{\text{as}}$ and $(\text{PO}_3^-)_{\text{s}}$ respectively [15,29].

4. Discussion

The reduction in density can be attributed to the lower molecular mass of CaF_2 (78.07 g/mol) when compared with BaO (153.33 g/mol). A small decreasing trend in molar volume may be due to the partial replacement of Ba^{2+} having larger ionic radius ($1.35\text{ \AA}$) with a smaller ionic radius ($1.0\text{ \AA}$) Ca^{2+} [17,34]. Reduction in the free volume (total interstitial space), caused by the substitution of Ba^{2+} having smaller field strength ($0.24 \times 10^{-20}\text{ m}^{-2}$) with Ca^{2+} having larger field strength ($0.35 \times 10^{-20}\text{ m}^{-2}$) leads to slight increase in the atomic packing density [13,17,35].

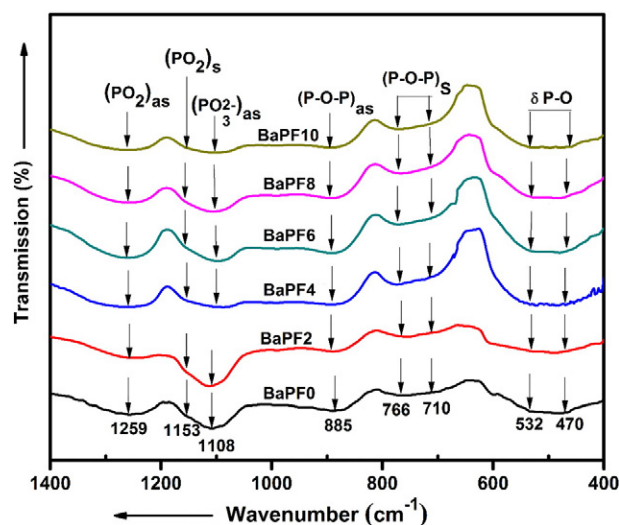


Fig. 4. FTIR spectra of BaPF glasses.

Table 5
Infrared absorption frequencies (cm^{-1}) and modes of vibration of BaPF glasses.

Characteristic groups and modes of vibration	Infrared absorption frequencies (cm^{-1})					
	BaPF0	BaPF2	BaPF4	BaPF6	BaPF8	BaPF10
Asymmetric stretching of PO_2 , $(\text{PO}_2)_{\text{as}}$ and stretching mode of $\text{P}=\text{O}$	1259	1258	1259	1259	1259	1260
Symmetric stretching of PO_2 , $(\text{PO}_2)_{\text{s}}$	1153	1153	1153	1153	1153	1153
Asymmetric stretching of PO_3^{2-} , $(\text{PO}_3^{2-})_{\text{as}}$	1108	1107	1101	1101	1101	1101
Asymmetric stretching of $\text{P}-\text{O}-\text{P}$, $(\text{P}-\text{O}-\text{P})_{\text{as}}$	885	890	890	891	895	895
Symmetric stretching of $\text{P}-\text{O}-\text{P}$, $(\text{P}-\text{O}-\text{P})_{\text{s}}$	766	766	768	772	772	774
	710	711	713	713	713	715
Motion of cations against PO_4^{3-} and bending vibrations of $\text{O}-\text{P}-\text{O}$ units, $\delta(\text{PO})$	532	532	531	532	532	532
	470	470	470	468	468	465

The substitution of BaO by CaF_2 in the glass batch results in increase in the calculated values of elastic moduli of glasses due to the higher dissociation energy (G_i) and packing factor (V_i) of CaF_2 compared to that of BaO. Poisson's ratio of glasses was found to remain almost constant due to the negligible variation of packing density of the samples (V_t in Table 1) with increase in CaF_2 content. Poisson ratio for the BaPF glasses calculated using Eq. (5) is 0.26, which is found to lie within the range of values reported for phosphate glasses ($\sigma > 0.25$) [22].

It can be observed from Table 3 that Vickers hardness of BaPF glasses increases with increase in CaF_2 content. This may be attributed to the increase in the strength of phosphate bonds caused by the partial substitution of Ba^{2+} with Ca^{2+} having higher field strength. Incorporation of calcium ions with smaller radius into the glass network also reduces the distance between anions and network modifier cations present in the glass, which results in strengthening of the glass network and increase in the hardness of the glasses [13]. Increase in the half crack length and increase in hardness result in a small reduction in fracture toughness. Increase in measured half crack length of glasses with high CaF_2 content indicates that cracks propagate easily through these samples. Propagation of the cracks in the glasses occurs due to the breakage of weakest bonds present in the sample, so easy propagation of the cracks through the glasses with high CaF_2 content may be due to the formation of a greater number of bonds between modifying cations and nonbridging oxygen. The strength of the bond between modifying cations and nonbridging oxygen are much weaker than the bonds between phosphorous and bridging oxygen or phosphorous and nonbridging oxygen [36]. Brittleness of glasses depends on densification and plastic flow modes of deformation before the crack initiation process. Increase in brittleness of BaPF (high density, $> 2.4 \text{ g/cm}^3$) glasses with increase in

CaF_2 content is due to lesser densification [24]. It can be observed from Table 4 that for both the samples, mass percentage of elements obtained from EDS analysis are in reasonable agreement with the calculated values from batch composition. EDS analysis helped to ascertain change in calcium content with CaF_2 addition in BaPF10 sample. Inability of EDS to detect the light elements (Be, B, C, N, O, F) present in low quantities may be the reason for not detecting fluorine present in BaPF10 sample. In our previous study [8], we have investigated the effect of CaF_2 addition on thermal and optical properties of these samples and found that both Ca^{2+} and F^- have effect on these properties, which confirmed the presence of fluorine in the prepared samples.

Slight shift of asymmetric and symmetric stretching of $\text{P}-\text{O}-\text{P}$ band to a higher wavenumber with increase in CaF_2 content observed in FTIR spectra may be due to change in length of the phosphate chain and the $\text{P}-\text{O}-\text{P}$ bond angle. The $\text{P}-\text{O}-\text{P}$ bond angle depends on the metal cations present in the phosphate glass structure [37]. Higher field strength and the smaller size of Ca^{2+} leads to smaller $\text{P}-\text{O}-\text{P}$ bond angle and results in shift of $\text{P}-\text{O}-\text{P}$ bands to a higher wavenumber. This band shift indicates the increase in covalent character of $\text{P}-\text{O}-\text{P}$ bond resulting in strengthening of $\text{P}-\text{O}-\text{P}$ bond as BaO is substituted by CaF_2 in the batch composition [38]. $(\text{PO}_3^{2-})_{\text{as}}$ band at $\sim 1108 \text{ cm}^{-1}$ shows a tendency for disappearance as the CaF_2 content increases, which may be attributed to the availability of more number of oxygen neighbors for Ca^{2+} ions to form its tetrahedra or octahedra compared to that of a Ba^{2+} ions [2,38]. Shift of $(\text{PO}_3^{2-})_{\text{as}}$ band to lower wavenumber with increase in CaF_2 content may be attributed to the higher ionic nature of phosphorous-oxygen bonds linked to calcium in $\text{P}-\text{O}-(\text{Ca})$ than that linked to barium in $\text{P}-\text{O}-(\text{Ba})$ bonds. Higher covalence nature of calcium-oxygen bond as compared to the barium-oxygen bond leads to the more ionic nature of phosphorous-oxygen bonds linked to calcium in $\text{P}-\text{O}-(\text{Ca})$ [38].

Raman spectra of BaPF glasses indicate that mainly the symmetric stretching mode of the two non-bridging oxygen atoms bonded to phosphorus atom, $(\text{PO}_2)_{\text{s}}$, and symmetric stretching vibrations of $\text{P}-\text{O}-\text{P}$ bridges of Q^2 units in the metaphosphate chain, $(\text{P}-\text{O}-\text{P})_{\text{s}}$, are affected by the substitution of BaO by CaF_2 in the batch composition. A similar trend is observed in the Raman spectra of mixed strontium-manganese metaphosphate glasses, mixed alkali metaphosphate glasses and mixed alkaline earth metaphosphate glasses studied earlier [29,39,40]. PO_2 and $\text{P}-\text{O}-\text{P}$ symmetric stretching vibration peaks shift to higher wavenumber as BaO is replaced by CaF_2 in the batch composition. This may be due to partial substitution of Ba^{2+} having larger ionic radius (1.35 Å) with a smaller ionic radius (1.0 Å) Ca^{2+} . This results in smaller $\text{O}-\text{P}-\text{O}$ and $\text{P}-\text{O}-\text{P}$ bond angles as bond angles are smaller for smaller cations in the alkali earth metal metaphosphate crystals [40]. Reduction in peak intensities of PO_2 symmetric vibration may be attributed to increase in covalent character of metal ion oxygen (M-O) bond, resulting in stronger coupling between metal cations and phosphate tetrahedra, thereby increasing the overall strength of phosphate network [15,40]. Gradual increase in wavenumber and reduction in peak intensities of PO_2 symmetric vibration with CaF_2 addition

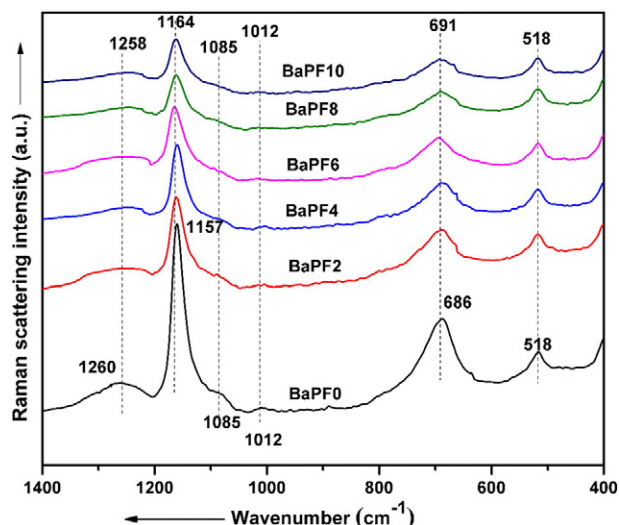


Fig. 5. Raman spectra of BaPF glasses.

indicate that this vibration is associated with an average cation environment rather than a particular cation domain environment [39]. Trivial shift of PO_2 asymmetric stretching vibration to lower wavenumber with increase in CaF_2 content may be due to decrease in length of phosphate chains [33].

Vibrational Raman spectra of BaPF glasses show only one band for each mode and it varies gradually with increase in CaF_2 content. It indicates that cations (Ba^{2+} & Ca^{2+}) are randomly distributed throughout the glasses and there is no molecular level ionic domain formation. Thus the phosphate chain vibration modes are associated with an averaged cation environment [39].

Infrared and Raman spectroscopy of prepared glasses indicate that vibrationally significant local geometry and forces associated with a particular cation are unaffected by the addition of second cation into the glass matrix. Cation dependent shift in wavenumber and change in width of bands are due to change in network bond angles and variation in the metal-oxygen force constant.

5. Conclusions

Physical properties such as density and molar volume show a decreasing trend while atomic packing density slightly increases with increase in CaF_2 content. Elastic parameters calculated using the Makishima and Mackenzie (M–M) model lies within the range of values reported for phosphate glasses. The mechanical properties like Vickers hardness and brittleness increase whereas the fracture toughness of the samples decreases as BaO content decreases. Analysis of radial-median crack generated by Vickers indentation on the surfaces of prepared glasses using scanning electron microscopy further confirms an increase in the brittleness of samples with increase in CaF_2 content. Changes in the physical and mechanical properties of ternary barium phosphate glasses, $(50 - X)\text{BaO}-X\text{CaF}_2-50\text{P}_2\text{O}_5$ with increase in CaF_2 content show that these properties depend on the type of cations present in the glass network. Interaction between cations (Ba^{2+} and Ca^{2+}) with network forming phosphorus tetrahedra result in structural changes and lead to change in these properties. Infrared and Raman spectroscopy reveal that the glass network mainly consists of metaphosphate (Q^2) units arranged in chain and ring structures together with small amount of ultraphosphate (Q^3) and pyrophosphate (Q^1) units.

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